

AERIS-F411

Advanced Embedded Research & Intelligent System
Flight Controller & Data Logger Development Board

Design reference document — Rev A

4-layer PCB, quadcopter-shaped outline, STM32F411RET6 core

This document is a complete design reference for building the schematic and layout in KiCad: mechanical outline, layer stack, bus assignments, full BOM with sourcing, sheet-by-sheet wiring specifications with component values, firmware bring-up order, and a pre-fabrication design verification checklist. It is not an exported KiCad schematic file — the actual symbol placement and wiring must be done in the KiCad schematic editor using the specifications below.

1. Mechanical outline

Quadcopter-shaped 4-layer PCB, sized to match a standard 5-inch frame hole spacing.

Parameter	Value
Overall diagonal span (hole to hole)	160 mm
Central body	46 mm x 46 mm square
Arm width	20 mm
Mounting holes	M3, 3.2 mm diameter, one per arm tip
Board thickness	1.6 mm standard
PCB layers	4

2. Layer stack

Layer	Assignment	Notes
L1 (top)	Components + signals	USB diff pair, SPI, I2C. Keep USB D+/D- length matched within 0.1mm
L2	Solid ground plane	Never split, even under the buck converter
L3	Power plane	5V and 3V3 as separate, moated copper pours
L4 (bottom)	Signals	GPIO, UART, slower signals. Keep buck converter switching loop small here

3. Bus assignment

Bus	Devices	Reasoning
SPI1	ICM-42688 IMU, W25Q64JV flash, microSD (separate CS heads)	ICM-42688 highest achievable sample rate
I2C1	BME280, DS3231M RTC	Both slow-pollled; shared bus is fine
UART2	GPS header	NMEA, needs dedicated UART
UART1	Debug console / USB CDC	Kept separate from GPS parsing

4. Component sourcing table

KiCad = ships in official kicad-symbols / kicad-footprints libraries. SnapEDA = not in KiCad defaults, export from snapeda.com using the listed search term.

Block	Part	Source	KiCad lib:symbol / footprint or search term
MCU	STM32F411RET6	KiCad	MCU_ST_STM32F4:STM32F411RETx -> LQFP-64_10x10mm_P0.5mm
HSE crystal	8 MHz	KiCad	Device:Crystal_GND24 -> Crystal_SMD_3225-4Pin_3.2x2.5mm
LSE crystal	32.768 kHz	KiCad	Device:Crystal_GND24_Small -> Crystal_SMD_3215-2Pin_3.2x1.5mm
USB-C connector	16-pin receptacle	KiCad	Connector:USB_C_Receptacle_USB2.0_16P -> USB_C_Receptacle_GCT_US
USB ESD	USBLC6-2SC6	KiCad	Diode:USBLC6-2SC6 -> SOT-23-6
IMU	ICM-42688-P	SnapEDA	search "ICM-42688-P"
Environmental sensor	BME280	SnapEDA	search "BME280"
SPI flash	W25Q64JVSSIQ	KiCad	Memory_Flash:W25Q64JVxxIQ -> SOIC-8_3.9x4.9mm_P1.27mm
microSD socket	Push-push	KiCad	Connector_Card:microSD_HC_Molex_104031
RTC	DS3231M	SnapEDA	search "DS3231MZ"
Backup battery holder	CR1220	KiCad	Battery.kicad_sym generic cell -> BatteryHolder_Keystone_1058
Buck converter	TPS563201	SnapEDA	search "TPS563201"
3.3V LDO	AMS1117-3.3	KiCad	Regulator_Linear:AMS1117-3.3 -> SOT-223-3_TabPin2
Reverse polarity FET	SI2333	SnapEDA	search "SI2333"
Fuse	1206 PPTC, 2A	KiCad	Device:Fuse -> Fuse_1206_3216Metric
TVS diode	SMBJ12A class	KiCad	Device:D_TVS -> D_SMB
Battery connector	XT30	SnapEDA	search "XT30PW-M"
Status LEDs	0603 LED	KiCad	Device:LED -> LED_0603_1608Metric
RGB status LED	Common cathode	KiCad	Device:LED_RGB -> LED_RGB_1210_3216Metric
Buttons	SMD tactile	KiCad	Switch:SW_Push -> SW_SPST_B3U-1000P
CAN transceiver	SN65HVD230	SnapEDA	search "SN65HVD230"
RS485 transceiver	MAX3485	SnapEDA	search "MAX3485"
Power monitor	INA219	SnapEDA	search "INA219"
Generic headers	2.54mm pin headers	KiCad	Connector_Generic:Conn_01xN -> PinHeader_2.54mm
SWD debug	Tag-Connect footprint	SnapEDA	search "TC2030-IDC-NL" (or plain 2x5 0.05in header)
ESC signal pads	Bare pads x4	KiCad	Connector_Generic:Conn_01x03

5. Power sheet

5.1 Input and protection

USB-C 5V in -> USBLC6-2SC6 ESD -> joins 5V bus.

XT30 battery in (2S LiPo, 7.0-8.4V) -> SI2333 reverse polarity FET -> 2A fuse -> SMBJ TVS diode -> TPS563201 buck converter (out: 5V) -> joins 5V bus via power-OR.

Component	Value	Purpose
Buck input caps	10uF X7R 25V x2	Input decoupling for 2S battery range
Buck inductor	4.7uH, Isat > 3A	Per TPS563201 typical application, 5V out
Buck output caps	22uF X7R 16V + 100nF	Output ripple filtering
Feedback R top (Rf _{bt})	20 kohm	Sets V _{out} with R _{fbb}
Feedback R bottom (R _{fbb})	6.8 kohm	Gives ~5.03V out with R _{fbt} above
Bootstrap cap	100nF	Standard, per datasheet
CC1/CC2 pulldowns (USB-C)	5.1 kohm to GND each	Signals 5V default-current sink
D+/D- series resistors	22 ohm	ESD/impedance matching
Reverse polarity FET gate R	10 kohm to GND	Conducts normally, blocks if battery reversed
Fuse	2A resettable PPTC	Rated above peak current draw
TVS diode	SMBJ12A class	Clamps transients above ~12V

5.2 Regulation and distribution

5V bus -> INA219 current monitor (I2C readout) -> AMS1117-3.3 LDO -> 3.3V rail, fanned out to MCU+SWD, IMU+BME280, flash+microSD, and RTC+coin cell.

Component	Value
LDO input cap	10uF
LDO output cap	22uF, needs adequate ESR (tantalum or ceramic + series R)

6. MCU core sheet

Sub-circuit	Component	Value
Power decoupling	Per VDD pin (x4 typical)	100nF ceramic, placed as close to pin as possible
Power decoupling	VDD bulk	4.7uF ceramic near MCU
Power decoupling	VDDA filter	Ferrite bead from VDD, then 100nF + 1uF at VDDA pin
Power decoupling	VBAT	Tie directly to VDD
Reset circuit	NRST filter cap	100nF to GND (internal pull-up already present)
Reset circuit	Reset button	SMD tactile switch, NRST to GND when pressed
Boot select	BOOT0 pulldown	10 kohm to GND (default: boot from flash)
Boot select	BOOT0 button	Momentary switch pulling BOOT0 to 3.3V for DFU bootloader
Oscillator	HSE load caps	2x 10pF (check exact crystal load capacitance spec)
Oscillator	LSE load caps	2x 6-9pF, typical for 32.768kHz watch crystal
SWD header pin 1	VDD	3.3V, probe voltage sense only
SWD header pin 2	SWDIO	-
SWD header pin 3	GND	-
SWD header pin 4	SWCLK	-
SWD header pin 5	NRST	-
SWD header pin 6	SWO (optional)	For trace/printf debugging

7. USB interface sheet

Uses the same USBLC6-2SC6 ESD protection, CC pulldowns, and D+/D- series resistors already specified in section 5.1. No additional components needed on this sheet.

8. Storage and sensors sheet

8.1 SPI1 bus: STM32 master, shared MOSI/MISO/SCK, three chip-selects (IMU, flash, microSD)

Component	Value / note
ICM-42688 decoupling	100nF ceramic at VDD
ICM-42688 CS pull-up	10 kohm to 3.3V
W25Q64JV decoupling	100nF
W25Q64JV CS pull-up	10 kohm to 3.3V
microSD decoupling	100nF + bulk 1uF (write current spikes)
microSD CS pull-up	10 kohm to 3.3V
SPI series resistors (optional)	22-33 ohm on MOSI/SCK if traces exceed ~30mm

8.2 I2C1 bus: STM32 master, shared SDA/SCL (BME280, DS3231M)

Component	Value
SDA pull-up	4.7 kohm to 3.3V
SCL pull-up	4.7 kohm to 3.3V
BME280 decoupling	100nF
BME280 address select	SDO to GND = 0x76, SDO to VDD = 0x77
DS3231M decoupling	100nF, plus separate 100nF at VBAT/coin-cell pin
DS3231M address	Fixed 0x68

9. Interfaces sheet (CAN, RS485, GPS, expansion)

Component	Value / note
SN65HVD230 decoupling	100nF at VCC
SN65HVD230 RS pin	Tie to GND for high-speed mode
CAN termination	120 ohm across CANH/CANL, populate only if this board is a bus end-node
MAX3485 decoupling	100nF
MAX3485 DE/RE pins	Tie together, drive from one MCU GPIO
RS485 termination	120 ohm across A/B, same end-of-bus caveat as CAN
GPS UART header	TX, RX, GND, 3.3V or 5V per module — no additional components
Expansion headers (GPIO/PWM)	Bare breakout, no components

10. Connectors and indicators sheet

Component	Value / note
ESC signal pads (x4)	Signal + GND only, 10 kohm pulldown per signal line
User button	SW_Push, 10 kohm pull-up to 3.3V, 100nF debounce cap
Power LED (green)	330 ohm series resistor ($V_f \sim 2V$, $\sim 5mA$ at 3.3V)
Boot LED	330 ohm series resistor, driven from spare GPIO
Status RGB LED	Red ~ 220 ohm, green ~ 330 ohm, blue ~ 330 ohm (verify vs LED datasheet)

11. Firmware bring-up order

Stage	Test
1	Power rails, no MCU populated or programmed — verify 5V and 3.3V with a multimeter
2	GPIO blink test — minimal firmware, one LED toggling
3	Clock tree — verify HSE and LSE oscillators are running
4	SWD debug + UART console working
5	SPI: bring up IMU, then flash, then microSD in that order
6	I2C: bring up BME280, then RTC
7	CAN/RS485 — loopback test between two boards, or scope/logic analyzer check
8	GPS and INA219 power monitor — read-a-value tests
9	Full data-logging integration — IMU + BME280 + RTC timestamp logged to flash and microSD

12. Design verification checklist (before sending Gerbers)

Check	What to look for
ERC	Unconnected pins, conflicting outputs, missing power flags
DRC	Trace/space violations, annular ring size on smallest vias
Footprint vs datasheet	Verify every SnapEDA part against the real manufacturer datasheet
3D model check	Run KiCad 3D viewer, check for physical collisions (connectors, mounting screws)
Power net check	5V/3V3 as separate non-overlapping pours; GND plane on L2 has no accidental splits
BOM sanity check	Every reference designator has footprint + 3D model, no duplicate designators

End of design reference. Build the schematic sheets in KiCad using the wiring and values above, then return for a placement/routing review once a first-pass layout exists.